

Time: 90 Minutes

Instructions: 1. Attempt all questions

2. Write your answers on your own paper

3. All necessary working must be shown

4. Marks will be deducted for careless or badly arranged work

**QUESTION 1 (22 marks)**

(a) Sketch each of the following curves on separate number planes

(i)  $y = x^2 - 4$

2

(ii)  $y = |x + 1|$

2

(iii)  $x^2 + y^2 = 4$

2

(iv)  $y = \frac{1}{x+2}$

2

(b) State whether each of the above curves are functions?

4

(c) State domain and range of

(i)  $y = 2^{-x}$  (ii)  $y = |x + 1|$  (iii)  $y = \frac{1}{x+2}$  (iv)  $x^2 + y^2 = 4$

8

(d) State the domain of  $y = \frac{1}{\sqrt{8-2x}}$ 

2

**QUESTION 2 (17 marks)**(a) If  $g(x) = x^2 - 4x$  find

(i)  $g(2)$

(ii)  $g(1-w)$

(iii) Find  $x$  if  $g(x) = -3$

6

(b) Given that  $f(x) = \begin{cases} 2x-4 & \text{if } x \geq 1 \\ x-3 & \text{if } -1 < x < 1 \\ -4x^2 & \text{if } x \leq -1 \end{cases}$

Find (i)  $f(2)$ 

(ii)  $f(0)$

(iii)  $f(-1)$

(iv) Sketch the function

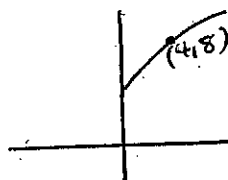
1

1

1

3

(c)

 $y=f(x)$  is an even function

(i) Redraw this function and complete the graph

2

(ii) Find  $f(4) + f(-4)$ 

1

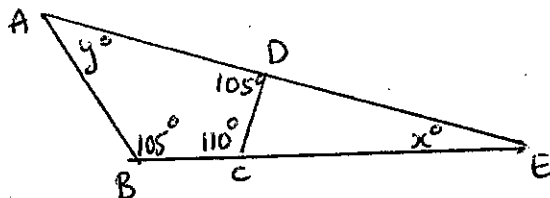
(d) Prove that  $f(x) = x^3 - 4x$  is an odd function

2

**QUESTION 3 (13 marks)**

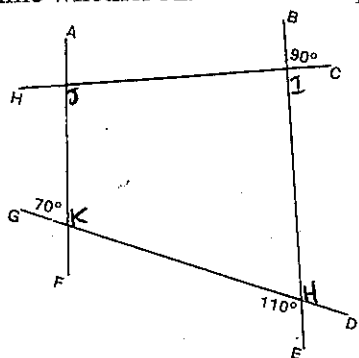
(a) Find the size of each interior angle of a regular octagon

2

(b) Find  $x$  and  $y$  giving reasons

4

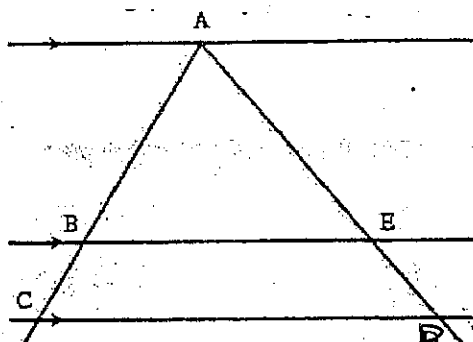
(c) Determine whether AF and BE are parallel with reasons



3

(d)

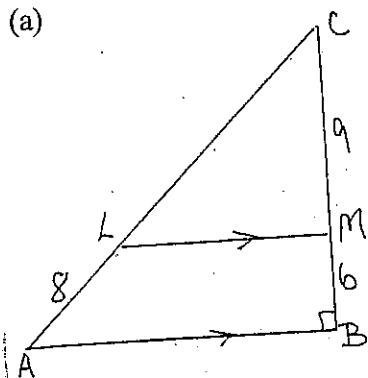
$AB=8\text{m}$ ,  $BC=3\text{m}$   $ED=5\text{m}$   
Find AD giving reasons



4

**QUESTION 4 (11 marks)**

(a)



(i) Find LC with reasons

2

(ii) Hence find AB as a surd

2

(b) A triangle has sides 12cm, 35cm, 37cm. Prove it is a right angled triangle.

2

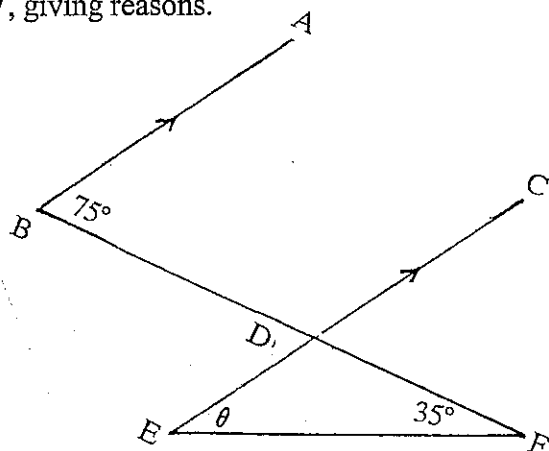
(c) A rhombus has perimeter 136cm and one diagonal is 60cm.

(i) Find the other diagonal as a surd

3

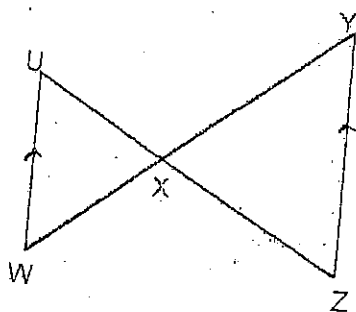
(ii) Find the area of the rhombus

2

**QUESTION 5 (17 marks)**(a) Find the size of  $\theta$ , giving reasons.

4

(b) The lines UZ and WY intersect at X as in the figure, UW is 10cm, WX is 8cm, UX is 7cm and XZ is 12cm.



(i) Draw a neat sketch and mark it with all the given information.

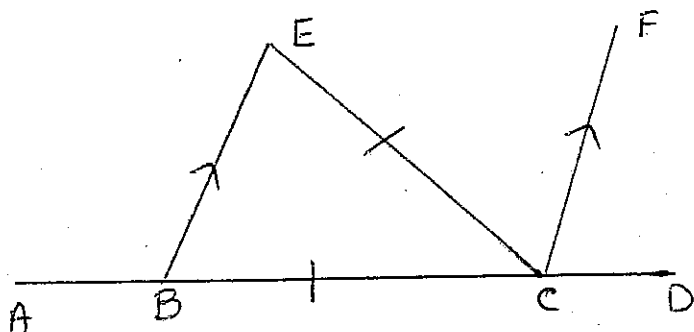
1

(ii) Prove that triangles UWX and XYZ are similar.

3

(iii) Hence or otherwise find the length of ZY with reasons

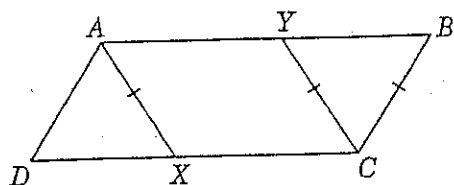
3

(c) ABCD is a straight line. Prove that CF bisects  $\angle ECD$  giving clear reasons

6

**QUESTION 6 ( 20 marks)**

(a) ABCD is a parallelogram. The point X lies on CD the point Y lies on AB and  $AX=CY=BC$  as shown in the diagram



(i) Copy the diagram onto your answer page

(ii) Explain why  $\angle ADX = \angle CBY$

1

(iii) Show that  $AD=AX$

3

(iv) Show that triangles ADX and CBY are congruent

4

(v) Hence prove that AYCX is a parallelogram

4

(b) ABCD is a rhombus, AX is perpendicular to BC and intersects BD at L

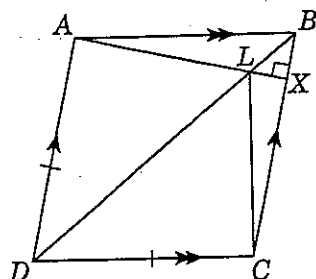


Figure not  
to scale

(i) Copy the diagram onto your answer paper and state why

$$\angle ADB = \angle CDB$$

1

(ii) Prove that the triangles ALD and CLD are congruent

4

(iii) Show that  $\angle DAL$  is a right angle with reasons

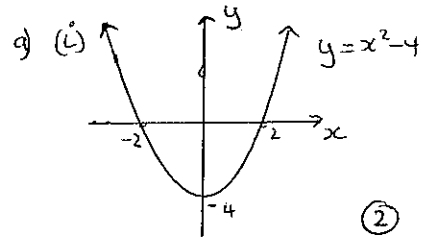
1

(iv) Hence or otherwise find the size of  $\angle LCD$  with reasons

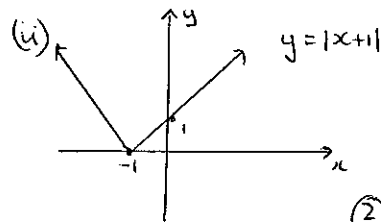
2

# Year 11 Mathematics Task 3

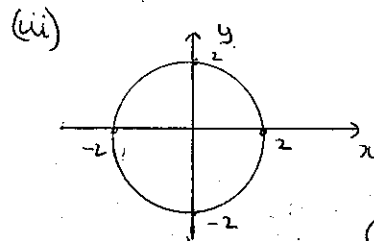
## Question 1



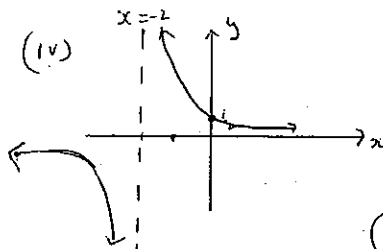
(2)



(2)



(2)



(2)

- (b) (i) function  
(ii) function  
(iii) not function  
(iv) function

(4)

## Question 1

(e) Domain:  $x$  all real  
Range:  $y > 0$

(2)

(ii) Domain:  $x$  all real  
Range:  $y \geq 0$

(2)

(iii) Domain: all real,  $x \neq -2$   
Range: all real,  $y \neq 0$

(iv) Domain:  $-2 \leq x \leq 2$   
Range:  $-2 \leq y \leq 2$

(d) Domain  $8 - 2x > 0$   
 $-2x > -8$   
 $x < 4$

(2)

## Question 2

(a) (i)  $g(2) = 4 - 8$   
 $= -4$

(2)

(ii)  $g(1-w) = (1-w)^2 - 4(1-w)$   
 $= 1 - 2w + w^2 - 4 + 4w$   
 $= -3 + 2w + w^2$

(2)

(iii)  $x^2 - 4x = -3$   
 $x^2 - 4x + 3 = 0$   
 $(x-3)(x-1) = 0$   
 $x = 1 \text{ or } 3$

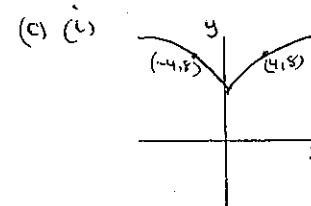
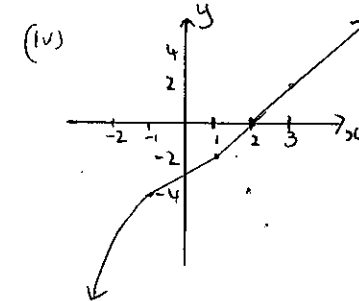
(2)

## Question 2

(b) (i)  $f(2) = 0$

(ii)  $f(0) = -3$

(iii)  $f(-1) = -4$



(ii)  $f(4) + f(-4) = 16$

(d)  $f(-x) = (-x)^3 - 4(-x)$   
 $= -x^3 + 4x$   
 $= -(x^3 - 4x)$   
 $= -f(x)$

(2)

## Question 3

(a)  $\frac{(8-2) \times 180}{8} = 135^\circ$

(2)

## Question 3

(b)  $y + 105 + 110 + 105 = 360$   
(angle sum of quad)  
 $y = 40^\circ$

(2)

$x + 40 + 105 = 180$   
(angle sum of triangle)

$x = 35^\circ$

(2)

(c)  $\angle IHK = 90^\circ$  (straight angle)

3

$\angle IHK = \angle JKG = 70^\circ$   
 $\therefore AF \parallel BE$  (corresponding angles are equal)

(d)  $\frac{8}{3} = \frac{x}{5}$  (ratio of intercepts)

2

$x = \frac{40}{3}$

$AD = \frac{40}{3} + 5$

$= 18\frac{1}{3} \text{ m}$

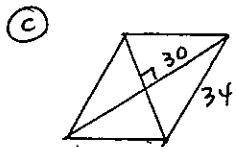
(4)

### Question 4

(a) (i)  $\frac{LC}{8} = \frac{9}{6}$  (ratio of intercepts)  
 $LC = 12$  (2)

(ii)  $AB^2 = 20^2 - 15^2$   
 $AB = \sqrt{175}$  (2)

(b) LHS =  $12^2 + 35^2$   
 $= 144 + 1225$   
 $= 1369$   
 $= 37^2$   
 $= RHS$   
 $\therefore$  triangles right angled

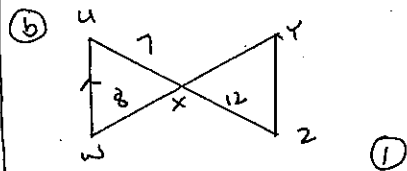


(i)  $x^2 = 34^2 - 30^2$   
 $= 256$   
 $x = 16$   
 $\therefore$  diagonal is 32cm

(ii)  $\frac{1}{2} \times 32 \times 60 = 960 \text{ cm}^2$

### Question 5

(a)  $\hat{CPB} = 105^\circ$  (co-interior angles  
 $AB \parallel CD$ )  
 $\hat{FDE} = 105^\circ$  (vertically opposite  
angle)  
 $\therefore \theta + 35 + 105 = 180$  (angle sum of  $\triangle$ )  
 $\therefore \theta = 40^\circ$



(i)  $\hat{YXZ} = \hat{UXW}$  (vertically  
opposite angles)  
 $\hat{WUX} = \hat{YZX}$  (alternate  
angles,  $uW \parallel YZ$ )  
 $\triangle uwx \parallel \triangle yxz$  (equiangular)

(3)

(ii)  $\frac{24}{10} = \frac{12}{7}$  (proportional  
sides of similar triangles)

$\therefore 24 = 17\frac{1}{7}$

(c) Let  $\hat{BEC} = x$

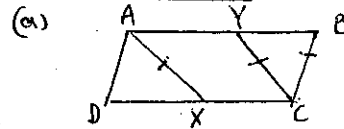
$\therefore \hat{ECF} = x$  (alternate  
angles  $BE \parallel CF$ )

$\hat{EBC} = x$  (base angles  
isosceles  $\triangle EBC$ ,  $EB = EC$ )

$\hat{FCD} = x$  (corresponding  
angles  $BE \parallel CF$ )

$\therefore \hat{ECF} = \hat{FCD} = x$   
 $\therefore CF$  bisects  $\hat{ECD}$

### Question 6



(i) opposite angles  
of parallelogram (1)

(ii)  $BC = AD$  (opposite  
sides of parallelogram)  
 $AX = BC$  (given)

$\therefore AX = AD = BC$  (3)

(iv)  $\hat{ADX} = \hat{YBC}$  (from part ii)

$\hat{AXD} = \hat{ADX}$  (angles  
opposite equal sides of  $\triangle$ )

$\hat{ABY} = \hat{CYB}$  (angles opposite  
equal sides of  $\triangle$ ).

$\therefore \hat{AXD} = \hat{CYB} = \hat{ABY}$

$CY = AX$  (given)

$\therefore \triangle ADX \cong \triangle CBY$  (AAS)

(4)

(vi)  $DX = BY$  (corresponding sides  
of congruent triangles)

$AB = DC$  (opposite sides of  
parallelogram)

$AB = AY + YB$

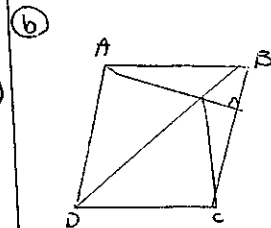
$DC = DX + XC$

$\therefore AY = XC = (AB - BY)$

$\therefore AX = YC$  (given)

$\triangle AXC \cong \triangle BYD$   
 $\therefore$  two pairs opposite sides equal

(4)



(i) In a rhombus diagonals  
bisect angle through which  
they pass.

(ii)  $AD = DC$  (equal sides of rhombus)  
 $\hat{ADB} = \hat{CDB}$  (from part i)

$DE$  common

$\triangle ADE \cong \triangle CDE$  (SAS) 4

(iii)  $\hat{DAE} = 90^\circ$  (alternate angles  
 $AD \parallel BC$ ) (1)

(iv)  $\hat{CED} = 90^\circ$  (corresponding  
angles of congruent  $\triangle$ 's).